

Lab 6: Matrix-Vector Multiplication

Math 124, Fall 2026 at the University of Michigan

due by the end of class on Wednesday, October 7, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, most of which will involve writing math on paper, and some of which will involve running code in a Jupyter Notebook. Lab activities are meant to last an hour, and the second hour of lab is dedicated to starting the homework assignment. To receive credit for a lab, you must show your lab TA your work on both the lab worksheet and homework assignment.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

This content is covered in [Chapter 3.1](#) and [Chapter 3.2](#) of the course notes, which are now posted. Have these open while working on the worksheet. Homework 5 will be released on Friday.

Activity 1: Is the expression valid?

Suppose:

- $A \in \mathbb{R}^{5 \times 3}$.
- $B \in \mathbb{R}^{3 \times 4}$.
- $\vec{u} \in \mathbb{R}^4$.
- $\vec{v} \in \mathbb{R}^3$.

Draw a  around each valid expression and ~~cross out~~ each invalid expression. An expression is valid if the shapes allow the operation shown. **Since you don't actually know A , B , $\vec{u}$, and $\vec{v}$, you can't actually do the calculation.**

$A\vec{v}$

$\vec{v}A$

$B\vec{u}$

$B\vec{v}$

$B\vec{u} + \vec{v}$

$A\vec{v} + \vec{u}$

$\vec{u} \cdot \vec{v}$

$\vec{v} \cdot \vec{v}$

$(B\vec{u}) \cdot \vec{v}$

$(A\vec{v}) \cdot (B\vec{u})$

$\|A\vec{v}\|^2$

$\|\vec{u} + \vec{v}\|$

$A(B\vec{u})$

$B\vec{u} - 3\vec{v}$

$\vec{u} + 2\vec{u}$

$\|B\vec{u}\|^2 + \|\vec{v}\|^2$

Compare with your group. Explain any disagreements using the shapes.

Activity 2: Fundamentals

Let

$$A = \begin{bmatrix} 4 & -3 & 6 \\ -2 & 5 & 1 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 \\ -3 & 4 \\ 5 & 2 \end{bmatrix}.$$

- a) Compute $A\vec{v}$. What are the shapes of $\vec{v}$ and $A\vec{v}$?

- b) Compute $B(A\vec{v})$. What is its shape?

- c) Is $B\vec{v}$ defined? Explain. Explain why $B(A\vec{v})$ can be defined even when $B\vec{v}$ is not.

Activity 3: Shuffle and stretch

Let

$$S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}, \quad \vec{x} = \begin{bmatrix} -4 \\ 7 \\ 9 \end{bmatrix}.$$

- a) Compute $S\vec{x}$ and $D\vec{x}$. Describe in words what each matrix does to the entries of an input vector.

- b) Compute $S(D\vec{x})$ and $D(S\vec{x})$.

- c) Why are $S(D\vec{x})$ and $D(S\vec{x})$ different? Explain using what S and D do to the entries of a vector.

Activity 4: Mystery matrices

There is a matrix A , but we have not been given its entries.

a) Suppose

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 12 \end{bmatrix}.$$

What does this tell you about A ?

b) We also know that

$$A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 1 \end{bmatrix}.$$

What does this tell you about A ? Use both pieces of information to write A .

c) There is another matrix B . We know only that

$$B \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 15 \\ 3 \\ 4 \\ 2 \end{bmatrix}, \quad B \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -6 \\ 11 \\ 0 \\ -5 \end{bmatrix}, \quad B \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -7 \\ 9 \\ 6 \end{bmatrix}.$$

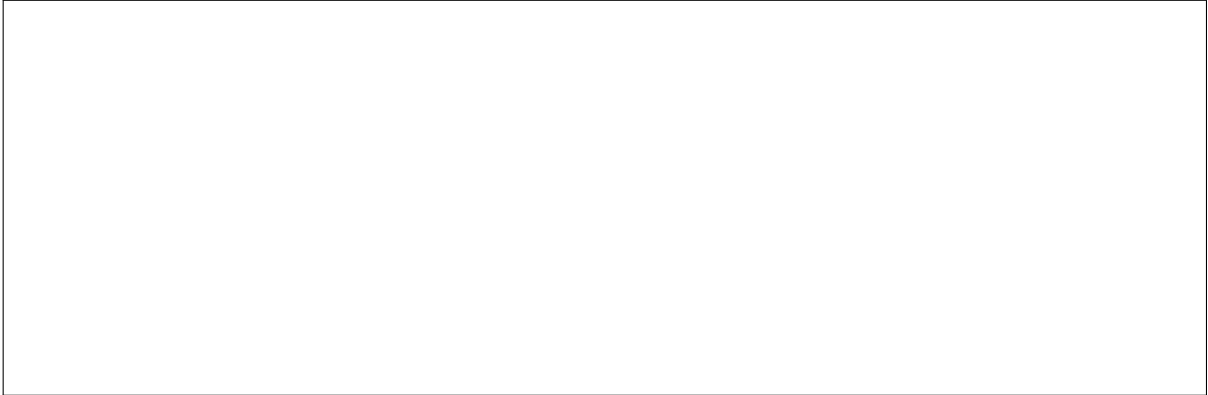
Using just this information, find $B \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}$. Explain how you used the three given outputs.

Activity 5: Drawing projections

Let $\ell = \text{span}(\vec{w})$, where

$$\vec{w} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}.$$

- a) First, let's refresh: find $\text{proj}_{\ell}(\vec{v})$ the usual way.



- b) We can also project onto ℓ by multiplying by a matrix. A **projection matrix** P satisfies $P\vec{v} = \text{proj}_{\ell}(\vec{v})$ for every input $\vec{v}$. If a unit vector $\vec{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ spans ℓ , then

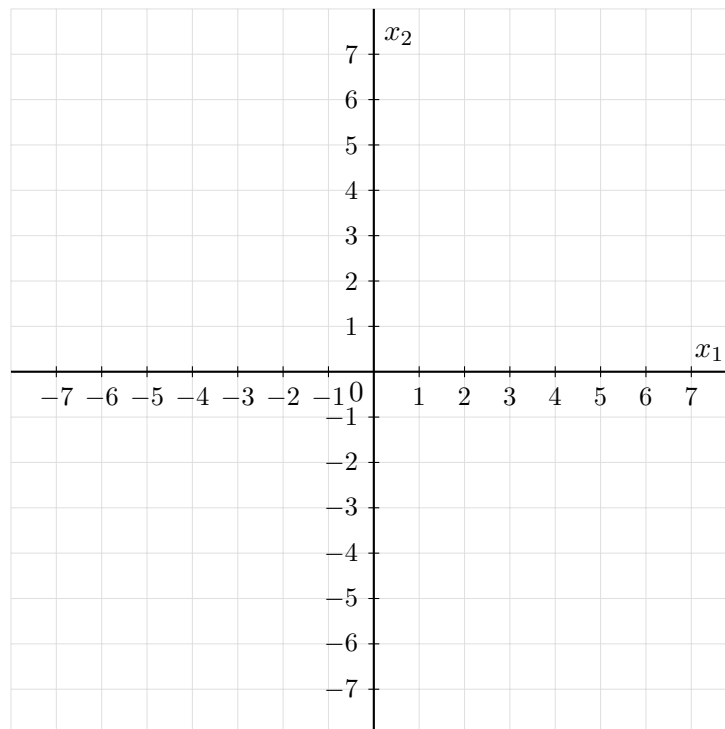
$$P = \begin{bmatrix} w_1^2 & w_1 w_2 \\ w_1 w_2 & w_2^2 \end{bmatrix}.$$

Find P for this line, then find $P\vec{v}$. Compare with your answer in part (a). What do you notice about the columns of P ?



Activity 5: Drawing projections (continued)

- c) Use P to find the projection onto ℓ of $\vec{u} = \begin{bmatrix} -4 \\ 5 \end{bmatrix}$. On the coordinate axes below, draw ℓ and the four vectors $\vec{u}$, $\vec{v}$, $P\vec{u}$, and $P\vec{v}$, all starting at the origin. What do you notice?



Activity 6: Programming: Matrix-Vector Multiplication

This lab has a programming notebook to help you visualize various matrix-vector operations, and to show you how matrix-vector multiplication works in Python.

To open the notebook for Lab 6, click the Google Colab link under “Code” on the course website for Lab 6. Instructions on how to use Google Colab are at math124.org/running-code. If you are viewing the lab worksheet after it has been posted, [this direct link](#) will work, too. To receive credit for the programming component of the lab, work through the entire notebook and show your lab TA your work.